Auditing Privacy Leakage in Tabular Foundation Model Embeddings

Xun Wang **Adam Dziedzic** **Michael Backes** **Franziska Boenisch**
CISPA Helmholtz Center for Information Security
{xun.wang, adam.dziedzic, backes, boenisch}@cispa.de

Abstract

Tabular Foundation Models (TabFMs) produce context-aware embeddings that are increasingly stored, shared, and reused for retrieval, clustering, and cross-institutional data exchange. These embeddings are often treated as privacy-preserving substitutes for raw tabular records, yet their attribute-level information content remains poorly understood. We present a systematic privacy audit of TabFM embeddings: given an embedding and varying amounts of side information, how accurately can sensitive attributes be recovered? We introduce *Cascade Probing*, a sequential probing method that measures recoverability while accounting for inter-attribute dependencies, significantly outperforming joint probing and optimization-based inversion baselines. Across four TabFMs (TabPFN, TabDPT, Mitra, TabICL) and ten datasets, we find that even without any side information, 65–95% of sensitive attributes can be recovered from embeddings alone. More critically, recoverability exhibits a *Privacy Cliff*: revealing a single known attribute sharply increases the recoverability of all remaining attributes. This phenomenon generalizes across model architectures and diverse domains, and is not eliminated by standard embedding transformations including noise injection, PCA, and knowledge distillation. These findings suggest that we need more formal privacy techniques for protecting sensitive information inside TabFM embeddings. Our code is available at `https://github.com/sprintml/PrivacyCliff`.

1 Introduction

Tabular Foundation Models (TabFMs) [16, 17, 24, 30] leverage in-context learning to achieve state-of-the-art performance on tabular prediction tasks without task-specific training. Beyond predictions, these models produce context-rich embeddings that are increasingly stored, shared, and reused as substitutes for raw tabular records, e.g., for retrieval, clustering, and cross-institutional data exchange [10, 28, 37, 41]. While these embeddings can be considered as abstractions of raw data, their attribute-level information content remains poorly understood. This raises a natural question: *how much sensitive information about the original record can be recovered from a TabFM embedding?*

We study this through *attribute inference*, which aims to recover the sensitive attribute values (e.g., income, diagnosis) of an individual record from the model's embedding output, given partial public knowledge. For classical tabular classifiers and regressors, prior work has shown that predictions alone can reveal hidden attributes of individual records [8, 9]. Unlike membership inference [32], which only asks whether a given data record was *present* during training, attribute inference directly measures what the representation *reveals about the record* it represents, making it well suited to assess the semantic privacy leakage in the setup of shared TabFM embeddings. In contrast to standard classifiers that initial attribute inference attacks have been designed for [8, 9], TabFMs differ in two ways. First, they expose high-dimensional *embeddings* rather than logits or class labels. Second, they natively support feature-level missing values, i.e., specific attributes can be masked with `NaN` while retaining others, and the resulting embedding is a well-defined model output. This enables

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differential queries that isolate the embedding signal attributable to masked attributes. Prior attribute inference methods, designed for predictions rather than embeddings with feature-level masking, may therefore underestimate the true leakage. We show this effect empirically in Section 5.2.

To overcome this limitation and to leverage the unique properties of TabFMs for stronger privacy auditing, we develop **Cascade Probing**, a method that recovers sensitive attributes sequentially rather than all at once. By decomposing the problem via the probabilistic chain rule and resolving easier attributes first, Cascade Probing exploits inter-attribute dependencies that simpler methods miss.

We apply Cascade Probing to audit four TabFMs (TabPFN [17], TabDPT [24], Mitra [43], TabICL [30]) in an evaluation spanning ten datasets, including four ACS Census benchmarks and two wider datasets with up to 100 features. Our audit reveals that leakage is severe across all models: Cascade Probing recovers 65–95% of sensitive attributes from embeddings alone, substantially more than existing inversion baselines. Models with higher-dimensional, more structured embeddings (TabDPT, TabICL) leak the most, while TabPFN's more entangled representations are harder to decode at baseline. Most critically, we discover a *Privacy Cliff*: knowing even a single public attribute of a record sharply increases recoverability of all remaining attributes, regardless of model or dataset. For practitioners, this means that sharing embeddings alongside even minimal metadata (e.g., demographic fields) can expose the entire record. Finally, we show that standard embedding-space mitigations (noise injection, PCA, knowledge distillation) fail to eliminate recoverability without destroying utility, indicating that privacy must be addressed at the representation level. Our main contributions are:

- We propose **Cascade Probing**, which leverages TabFMs' native NaN-masking to sequentially recover sensitive attributes via the chain rule. It recovers 65–95% of attributes from embeddings alone, outperforming all existing inversion baselines by large margins.
- We discover the **Privacy Cliff**: a single known attribute collapses the privacy of the entire record. This is intrinsic to how TabFMs encode information, not an artifact of the probing method.
- We provide the first systematic privacy audit across four TabFM architectures and ten datasets, showing that the vulnerability is systemic across domains and resists standard embedding-space mitigations.

2 Background

Tabular Foundation Models (TabFMs) are large-scale neural networks that generalize across diverse tabular datasets in a zero-shot manner via in-context learning (ICL). Given a support set $\mathcal{D}_{\text{sup}}$ of labeled examples at inference time, a TabFM produces a context-aware embedding $z = \mathcal{T}(x \mid \mathcal{D}_{\text{sup}})$ for each query record x before making predictions. Prior-Data Fitted Networks (PFNs) such as TabPFN [11, 16, 17] are trained on millions of synthetic datasets generated from structural causal models, effectively approximating Bayesian inference. Other architectures take different approaches: TabICL [30] uses a two-stage attention mechanism to scale to 500K samples, TabDPT [24] combines ICL with retrieval-based pre-training on real-world data, and Mitra [43] targets scalable deployment through AutoGluon. TabArena [7] demonstrates strong predictive performance of tabular foundation models on smaller datasets. Increasingly, users receive intermediate ICL embeddings z for tasks such as retrieval and clustering, rather than only final predictions. Recent analysis [40] reveals that these embeddings are highly separable and linearly decodable. While separability drives predictive performance, prior work on other modalities has shown that structured latent spaces can inadvertently expose sensitive attributes [23, 35]. Whether this risk applies to TabFM embeddings has not been studied and our work provides the first such audit.

Attribute Inference [8, 20] aims to reconstruct individual private attributes of data records from partial information, in contrast to full data reconstruction attacks typically studied in the image or language domain. The risk of attribute inference has been demonstrated across diverse settings, from social network analysis [21, 22, 45] to deep learning models [19, 36]. Our work studies this risk in a new modality: we measure recoverability from TabFM embeddings and optional side information, probing whether sensitive attribute values can be decoded from z. A comparison with other privacy risks, including membership inference and data extraction, is provided in Appendix D. Attribute inference is particularly relevant here because TabFM embeddings represent individual inference-time records, and the concrete risk of sharing them lies in the exposure of specific sensitive attribute values.

Embedding Inversion has been studied in NLP, where adversaries can recover text from sentence embeddings [34], with methods like Vec2Text [26] achieving high-fidelity reconstruction. This risk remains unexplored for TabFMs. Unlike holistic text reconstruction, the tabular domain requires recovering specific, heterogeneous attribute values. We address this gap by combining attribute inference with embedding inversion: our audit measures how accurately sensitive attribute values (x_{sens}) can be decoded from z and optionally known features (x_{pub}).

Tabular Data Synthesis. To enable auditing under data-scarce conditions, we leverage LLM-based tabular synthesis [3, 12, 33]. Building on few-shot tabular prompting [14] and chain-of-thought reasoning [38], we generate synthetic reference datasets from schema metadata and a small number of real samples to train the probing decoders (Section 4.5).

3 Auditing Attribute Leakage in TabFM Embeddings

As TabFM embeddings are increasingly shared for downstream use [28, 41, 43], auditing their privacy leakage is essential. Our goal is not to model a specific external adversary but to evaluate whether these embeddings are safe to share as data surrogates. Below, we formalize the audit objective and specify the assumptions under which recoverability is measured.

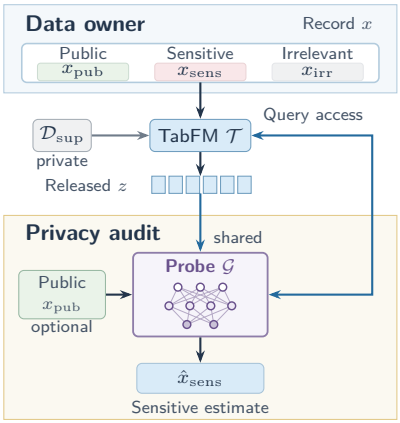


Figure 1: **Audit setup.** The auditor probes sensitive attributes using the released embedding, optional public attributes, and queries to the same TabFM. Support data remain private.

3.1 Audit Objective

We formulate the audit as a **conditional attribute recoverability** problem (Figure 1). A pre-trained TabFM $\mathcal{T}$ processes a query record x conditioned on a private support set $\mathcal{D}_{\text{sup}}$, producing a context-aware embedding $z = \mathcal{T}(x \mid \mathcal{D}_{\text{sup}})$. Each record x is partitioned into three disjoint sets: *public attributes* x_{pub}, which a recipient may plausibly know (e.g., Age, Race) and which serve as optional side information; *sensitive attributes* x_{sens}, whose recoverability we measure (e.g., Income, Diagnosis); and *irrelevant attributes* x_{irr}, the remaining features (e.g., Timestamp) masked with `NaN` during the audit. The set x_{pub} may be empty.

We estimate recoverability by learning a probing function $\mathcal{G}$ that approximates the maximum a posteriori (MAP) estimate:

$$\hat{x}_{\text{sens}} = \mathcal{G}(z, x_{\text{pub}}) \approx \underset{x_{\text{sens}}}{\operatorname{argmax}}\, P(x_{\text{sens}} \mid z, x_{\text{pub}}) \tag{1}$$

When $x_{\text{pub}} \neq \emptyset$, the probe exploits correlations between the side information and the embedding. When $x_{\text{pub}} = \emptyset$, recoverability is measured solely from z. High recoverability indicates that the embedding is unsafe to release without additional privacy protection.

3.2 Audit Assumptions

We parameterize the audit along three dimensions of information that may be available to parties who interact with the embedding downstream.

1. Access & Metadata. We assume **black-box query access to the same embedding endpoint** that produced the released embeddings. Crafted queries use the same TabFM and private support context, while model parameters and the private support set remain inaccessible. This evaluates what an internal auditor or an embedding recipient with such API access can recover. This setting reflects deployments in which users receive record embeddings and can submit additional records to the same embedding API [10, 28]. A static embedding release without access to the originating endpoint is outside our assumed access model. The auditor additionally possesses **metadata knowledge** of the dataset schema (e.g., column names, types). This is a minimal assumption: schema metadata is required to construct valid queries to the API in the first place, and is therefore available to any party with query access.

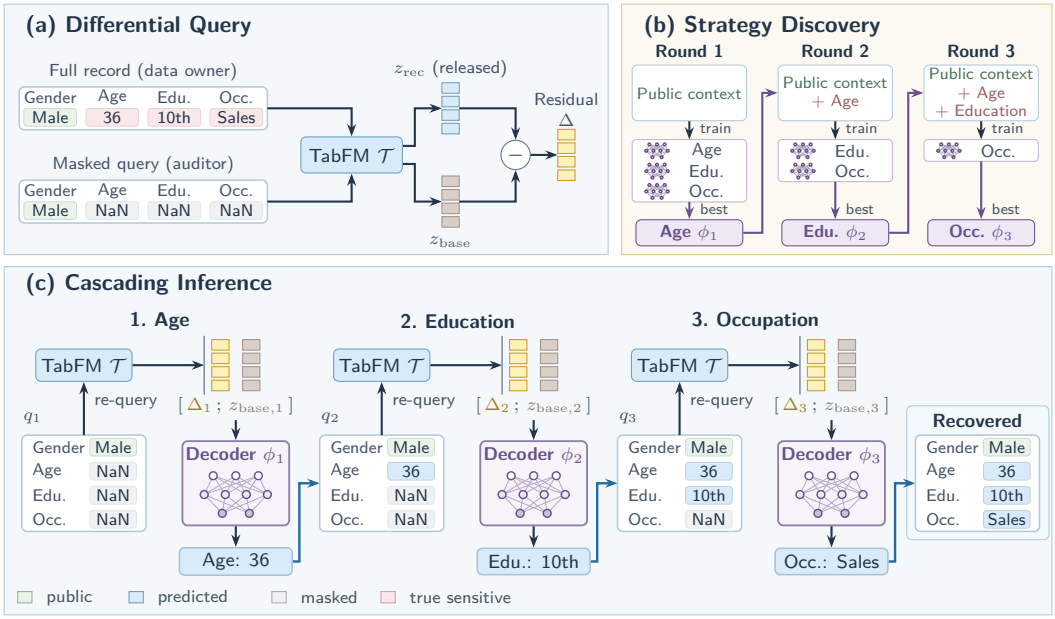


Figure 2: **Overview of Cascade Probing.** (a) Differential Query computes a baseline embedding from a masked query and subtracts it from the released embedding. (b) Strategy Discovery selects the best decoder on held-out reference data at each round, adding the selected attribute's true value to the next round's context. (c) Cascading Inference uses predicted values to update each query. The TabFM is queried again at every step, and the selected decoder receives the updated residual and baseline embedding. The released embedding stays fixed. The figure abbreviates the selected decoder ϕ_{π_t} as ϕ_t. Attribute values and recovery order are illustrative.

2. Side Information Level (K). We define the amount of side information about a specific record by $K = |x_{\text{pub}}|$, distinguishing two settings: **1) No-Context ($K = 0$):** The recipient knows nothing about the record ($x_{\text{pub}} = \emptyset$). Recoverability depends solely on the embedding. **2) Partial-Knowledge ($K \geq 1$):** The recipient knows at least one attribute value. This models realistic scenarios where some demographic or contextual information is available alongside the embedding.

3. Distribution Knowledge. We consider a spectrum from no knowledge to full knowledge of the data distribution, simulating what different recipients could achieve: **1) Shadow (full knowledge):** A recipient with access to a large same-distribution dataset ($\mathcal{D}_{\text{shadow}}$). This represents the strongest recipient and serves as an upper bound on recoverability. **2) Few-Shot (limited knowledge):** A recipient with only a handful of real samples (e.g., $N \leq 30$), used as seeds for LLM-based synthetic data generation. **3) Zero-Shot (no knowledge):** A recipient with no real data, relying on LLM-generated synthetic data from schema metadata alone.

4 Cascade Probing

The goal of Cascade Probing is to measure how accurately sensitive attributes of a record can be recovered from its TabFM embedding. Our key insight is that TabFMs encode strong feature dependencies; instead of predicting all sensitive attributes simultaneously, we probe them sequentially. By resolving more recoverable attributes first, we create a cleaner conditioning signal for probing harder ones. Our method, illustrated in Figure 2, consists of three stages: (1) **Differential Query**, (2) **Strategy Discovery**, and (3) **Cascading Inference**.

4.1 Differential Query

We leverage the native missing-value support of TabFMs to construct a reference embedding that separates public from sensitive information (Figure 2(a)). Let x be the target record. We construct

a masked baseline x_{base} by retaining the known public attributes x_{pub} while masking all sensitive attributes x_{sens} with `NaN`: $x_{\text{base}} = [x_{\text{pub}}, \texttt{NaN}, \ldots, \texttt{NaN}]$.

We query the model to obtain the baseline embedding $z_{\text{base}} = \mathcal{T}(x_{\text{base}})$. The **Embedding Residual** Δ is defined as the difference between the full record embedding z_{rec} and this baseline:

$$\Delta = z_{\text{rec}} - z_{\text{base}}. \tag{2}$$

The raw embedding z_{rec} is a composite representation encoding both public and sensitive features. By computing z_{base} as a reference, we obtain a differential view: the residual Δ captures the embedding shift induced by the sensitive attributes, while z_{base} retains the public context. We provide both as the concatenated input $[\Delta; z_{\text{base}}]$ to our probing models. Note that this is informationally equivalent to $[z_{\text{rec}}; z_{\text{base}}]$; the key enabler is the *differential query* itself, computing a reference embedding via native NaN-masking, which creates a baseline against which the sensitive signal can be measured.

4.2 Chain-Rule Decomposition

Tabular features exhibit strong dependencies. To exploit these dependencies, our **Cascade Probing** predicts sensitive attributes *one at a time*, conditioning each prediction on previously recovered attributes.

Joint Probing. As a baseline, Joint Probing masks all sensitive attributes and trains a single multi-task **probing decoder** to predict all sensitive targets simultaneously from the embedding representation.

Cascade Probing. Let $x_{\text{sens}} = \{a_1, \ldots, a_S\}$ be the set of S sensitive attributes and let $\mathcal{C} = \{z_{\text{rec}}, x_{\text{pub}}\}$ denote the initial information available to the auditor. By the probabilistic chain rule [2], the joint posterior can be factorized as

$$P(x_{\text{sens}} \mid \mathcal{C}) = \prod_{t=1}^{S} P(a_{\pi_t} \mid \mathcal{C}, \underbrace{a_{\pi_1}, \ldots, a_{\pi_{t-1}}}_{\text{Inferred Context}})$$

where π is a recovery order. This formulation motivates an auto-regressive probing procedure: at step t, we apply a per-attribute **probing decoder** ϕ_{π_t} that takes the residual and baseline embedding as input. Previously recovered values enter through the TabFM query used to compute these embeddings.

4.3 Strategy Discovery

The efficacy of the cascade depends critically on the recovery order π. Incorrect early predictions can propagate and amplify. To mitigate error propagation, we determine π offline via a greedy procedure on the auditor's reference dataset, prior to inference (Figure 2(b)).

Finding the globally optimal permutation is computationally intractable ($O(S!)$). We approximate it greedily by selecting the locally best attribute at each step ($O(S^2)$). As detailed in Algorithm 3 (Appendix E), at step t, given the set of previously selected attributes $\mathcal{R}$, we train a temporary probing decoder for every remaining attribute, conditioning on ground-truth values of $\mathcal{R}$ from the reference dataset. We evaluate each probe on a held-out validation set. In particular, the attribute with the best validation performance is selected as π_t and its ground-truth value is added to $\mathcal{R}$ for the next iteration. The selected decoder is retained for inference. This ensures the cascade recovers easier attributes first, building a stable context for harder ones.

4.4 Cascading Inference

At inference time (Figure 2(c)), ground-truth labels are unavailable, so the cascade uses its own predictions. At each step t, the auditor constructs a query q_t from x_{pub} and previously predicted values, leaving the remaining attributes masked. Querying the same TabFM endpoint gives $z_{\text{base},t} = \mathcal{T}(q_t)$ and the residual $\Delta_t = z_{\text{rec}} - z_{\text{base},t}$. The decoder ϕ_{π_t} retained during Strategy Discovery receives $[\Delta_t; z_{\text{base},t}]$ to predict the next attribute. The released embedding z_{rec} remains fixed throughout. Each prediction is locked into the context for the following step. Pseudocode for both Strategy Discovery and Cascading Inference is provided in Appendix E.

4.5 Implementation Details

Mixed feature types. Each probing decoder uses a classification head (Softmax + Cross-Entropy) for categorical attributes and a regression head for continuous ones. Continuous targets are min-max normalized to $[-1, 1]$ using bounds estimated from the reference dataset to prevent scale-dominant features from dominating training. During cascading inference, continuous predictions are inverse-transformed to their original scales before being fed back to the TabFM as context for subsequent steps. Details and ablations are in Appendix J.7.

LLM-based synthetic data. When a large reference dataset is unavailable, we synthesize $\mathcal{D}_{\text{syn}}$ via an LLM from schema metadata and a few real samples, using few-shot in-context learning, stratified conditional generation, and chain-of-thought prompting. $\mathcal{D}_{\text{syn}}$ is generated offline and used only to train the probing strategy (π, Φ).

5 Empirical Evaluation

We break our systematic privacy audit of TabFM embeddings down into five research questions: **RQ1:** How much attribute information is recoverable from TabFM embeddings? **RQ2:** Does partial side information sharply increase recoverability (the Privacy Cliff)? **RQ3:** Is the Privacy Cliff model-specific? **RQ4:** Is recoverability merely tabular correlation? **RQ5:** Can embedding transformations mitigate recoverability? We first describe the experimental setup, then address each RQ in turn.

5.1 Experimental Setup

Datasets. We evaluate on ten tabular datasets spanning diverse domains and scales. Four standard benchmarks serve as our primary evaluation suite: **Adult Census Income** [1], **German Credit** [15], **Heart Disease** [18], and **Diabetes Pima** [27]. We additionally evaluate on four ACS Census datasets from the folktables benchmark [6] to validate generalization across domains and feature compositions: **ACS Employment**, **ACS Public Coverage**, **ACS Income-to-Poverty**, and **ACS Health Insurance**. Each ACS dataset is subsampled to 50K records; TabPFN's default inference configuration further limits the support set $\mathcal{D}_{\text{sup}}$ to 10K samples. We further evaluate recovery on wider tables using COIL2000 [29] and Communities & Crime [31], with 85 and 100 features, respectively. Full dataset details are provided in Appendix B.

Evaluation Scenarios & Data Regimes. We evaluate attribute recoverability under varying side information and data access. In **No-Context** ($K = 0$), no side information is available ($x_{\text{pub}} = \emptyset$) and recoverability is measured solely from the embedding; in **Partial-Knowledge** ($K \geq 1$), we incrementally reveal public features (e.g., Race/Sex) to identify where recoverability saturates. Following Section 3, we consider three distribution knowledge regimes: **Shadow** (access to a same-distribution reference dataset comprising 50% of samples), **Few-Shot** (30 real samples to seed LLM synthesis), and **Zero-Shot** (metadata-only, LLM priors). For Few-Shot and Zero-Shot, we generate 2000 synthetic samples to train the probing decoders.

Splits and evaluation. We partition real records into disjoint support (40%), reference (50%), and test (10%) sets, subject to the support-size cap above. The reference pool in Shadow, or the synthetic pool in Few-Shot/Zero-Shot, is split 80/20 for decoder training/validation. All three regimes use the same held-out test records. We average results over three random seeds for each (K, S).

Attribute Partitioning. For each dataset, we designate a fixed set of *sensitive attributes* (x_{sens}) as recovery targets and a set of *public attributes* (x_{pub}) as potential side information, based on domain-informed priority lists. For example, on Adult, sensitive attributes include Occupation, Age, and Education, while public attributes include Race and Sex. Full priority lists and partitioning details for all datasets are in Appendix C.

Metrics. We report mean categorical accuracy (↑) for categorical features and Normalized Mean Absolute Error (NMAE, ↓, normalized to $[0, 1]$) for continuous features. We additionally report balanced accuracy and macro-F1 (mean class-wise F1 score), giving each class equal weight.

Models. We use TabPFN v2.5 [11] (192-dim embeddings) as the primary model under audit and validate cross-model generalizability on three additional TabFMs: TabDPT [24] (768-dim), Mitra [43] (512-dim), and TabICL [30] (512-dim). For synthetic data generation, we utilize Qwen2.5-7B-Instruct

Table 1: Mean categorical accuracy (↑) at $K=0$ (Shadow setting), averaged over all S. Best per row in bold. NMAE results in Appendix H.1.

Dataset	Joint	DirectOpt	WGAN	TabSyn	UIFV	Cascade
Adult Census Income	0.220	0.460	0.590	0.550	0.629	**0.650**
German Credit	0.440	0.540	0.620	0.610	0.499	**0.740**
Heart Disease	0.550	0.690	0.750	0.750	0.671	**0.790**
ACS Employment	0.517	0.620	0.620	0.680	0.882	**0.900**
ACS Public Coverage	0.460	0.450	0.440	0.420	0.673	**0.750**
ACS Income-to-Poverty	0.332	0.530	0.530	0.510	0.839	**0.845**
ACS Health Insurance	0.419	0.450	0.430	0.430	0.621	**0.686**

as the generator. Implementation details including decoder architecture and training hyperparameters are provided in Appendix F.

Decoder architecture. Probes use a three-layer MLP with hidden widths $[512, 256, 128]$, ReLU activations, and dropout 0.1. Joint Probing shares a backbone across target heads; Cascade Probing uses an independent decoder per step. We train with AdamW (learning rate 10^{-3}, batch size 64) for up to 100 epochs with early-stopping patience 10. Full hyperparameters are in Appendix F.

Baselines. We compare Cascade Probing against five baselines. (1) **Joint Probing** predicts all sensitive attributes simultaneously using a single multi-task decoder. (2) **Direct Optimization** [8] reconstructs attributes by gradient descent on the embedding distance. (3) **GMI+WGAN-GP** [13, 44] combines generative model inversion (GMI) with a tabular WGAN-GP prior. (4) **GMI+TabSyn VAE** [42, 44] replaces the GAN prior with the VAE component of TabSyn. (5) **UIFV Query Attack** [39] learns an inverse network to reconstruct attributes from intermediate representations using query access. We evaluate it on TabFM embeddings in the Shadow setting. Baselines without target embeddings are described in Section 5.5. Iterative refinement is evaluated in Appendix J.4.

5.2 RQ1: How Much Attribute Information Is Recoverable from TabFM Embeddings?

Motivation. We first establish the baseline privacy risk: how much can be recovered from a TabFM embedding when the auditor has no side information about the record ($K=0$)? This represents the minimum leakage inherent to the embedding itself.

Summary of Findings. On TabPFN v2.5, Cascade Probing achieves 65–90% categorical accuracy across the datasets with categorical features, achieving higher accuracy than all evaluated baselines. The gap is largest on complex, high-dimensional datasets (e.g., +51 percentage points over Joint Probing on ACS Income-to-Poverty), confirming that sequential conditional probing exploits inter-attribute dependencies that joint methods miss for risk reporting.

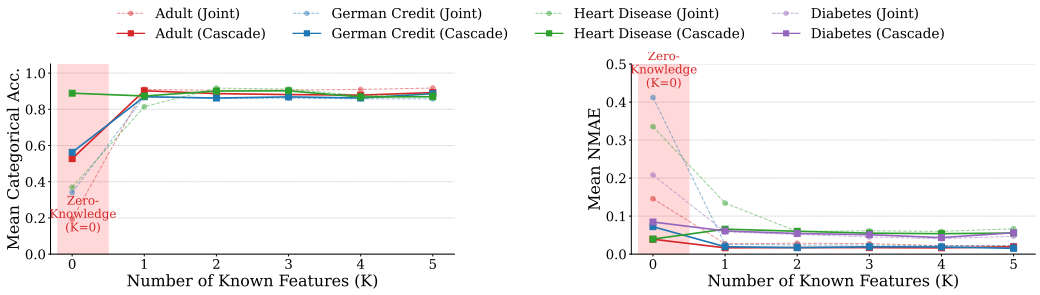
Detailed Results. Table 1 reports No-Context ($K=0$) recoverability in the Shadow setting. Cascade Probing outperforms all baselines on every dataset. Compared with UIFV, the gains range from 0.6 percentage points on ACS Income-to-Poverty to 24.1 points on German Credit. Optimization-based methods (DirectOpt, WGAN, TabSyn) improve over Joint Probing but remain substantially below Cascade, as they attempt to invert the entire embedding jointly without modeling inter-attribute dependencies. Diabetes Pima (all continuous) is reported via NMAE in Appendix H.1. Notably, even Joint Probing recovers 22–55% of attributes, indicating that TabFM embeddings encode substantial attribute information even under the simplest probing strategy. We further investigate whether this recoverability stems from the embedding itself rather than tabular correlations in Section 5.5.

Per-attribute recovery. Additional metrics show that recovery is not solely explained by class imbalance (Table 31). At $K=0$, Adult education achieves 0.65 balanced accuracy and 0.57 macro-F1 across 16 classes, compared with a constant predictor's balanced accuracy of $1/16 \approx 0.06$. German checking-account status achieves 0.98 balanced accuracy and 0.97 macro-F1 across four classes. Gains from $K=0$ to $K=1$ also remain evident in balanced accuracy and macro-F1.

Table 2 further shows that Cascade Probing yields effective audits under limited data access. Few-Shot (30 real samples + LLM synthesis) retains most Shadow-level recoverability, and on Heart Disease even exceeds Shadow (89% vs. 79%). Zero-Shot (no real data) achieves 49–88% accuracy,

Table 2: Cascade Probing at $K=0$ across data regimes, averaged over all S. Acc (↑) for categorical, NMAE (↓) for continuous features.

Dataset	Acc (↑)			NMAE (↓)		
	Shadow	Few-Shot	Zero-Shot	Shadow	Few-Shot	Zero-Shot
Adult	0.650	0.550	0.530	0.030	0.040	0.040
German	0.740	0.590	0.490	0.050	0.070	0.070
Heart	0.790	0.890	0.880	0.110	0.040	0.040
Diabetes	N/A	N/A	N/A	0.080	0.080	0.090



(a) **Categorical Features** (Higher is better). (b) **Continuous Features** (Lower is better).

Figure 3: **The Privacy Cliff.** Attribute recoverability as a function of side information (K known features) in the Few-Shot setting. The red shaded region highlights the No-Context ($K=0$) regime. Both Joint (dashed) and Cascade (solid) drastically jump at $K=1$; recoverability saturates for $K>1$.

comparable to Few-Shot, confirming that schema metadata alone enables practical auditing. Together, these results show that the practical privacy risk does not require access to a large reference dataset.

5.3 RQ2: Does Partial Side Information Sharply Increase Recoverability?

Motivation. In practice, a recipient of shared embeddings may also possess some public attributes of the individuals (e.g., demographic information). We investigate whether even a small amount of side information fundamentally changes recoverability.

Summary of Findings. A single known attribute triggers a drastic jump in recoverability across all datasets, for both Joint and Cascade Probing. This Privacy Cliff is intrinsic to the embeddings, and shows a sharp rise in recoverability at $K=1$ followed by saturation as further attributes are revealed.

Detailed Results. Figure 3 shows recoverability as side information increases from $K=0$ to $K=5$ (Few-Shot setting). At $K=1$, Joint surges from 21% to 91% on Adult and from 33% to 87% on German Credit; Cascade shows a similar jump on these datasets (55%→90% and 59%→87%, respectively). That even the simpler Joint method recovers nearly all attributes with one known feature confirms the cliff is a feature of the embedding, not an artifact of the probing method. Recoverability saturates for $K>1$, and the cliff appears regardless of which single attribute is revealed (Appendix J.3). This suggests that TabFM embeddings are highly **entangled**: knowing a single feature sharply reduces uncertainty over the remaining attributes, creating a binary privacy regime that shifts from substantial leakage at $K=0$ to broad exposure of remaining attributes at $K\geq 1$.

5.4 RQ3: Is the Privacy Cliff Model-Specific?

Motivation. The results above are obtained primarily on TabPFN. To determine whether the Privacy Cliff is a systemic property of TabFM representations rather than an artifact of a specific architecture, we evaluate Cascade Probing on three additional TabFMs with different embedding dimensions and training procedures.

Summary of Findings. All four TabFMs exhibit high recoverability and a clear $K=0 \rightarrow K=1$ jump, with the most pronounced cliff on TabPFN. The Privacy Cliff is not specific to TabPFN, but a shared property of TabFM representations.

Table 3: Cross-model generalization (Few-Shot, Cascade Probing). Accuracy (↑) for categorical features; NMAE (↓) for Diabetes Pima. The $K{=}0 \rightarrow K{=}1$ jump confirms the Privacy Cliff across all TabFMs.

	TabPFN		**TabDPT**		**Mitra**		**TabICL**	
Dataset	$K{=}0$	$K{=}1$	$K{=}0$	$K{=}1$	$K{=}0$	$K{=}1$	$K{=}0$	$K{=}1$
Adult (Acc)	0.550	0.900	0.930	0.960	0.860	0.900	0.950	0.970
German (Acc)	0.590	0.870	0.890	0.970	0.780	0.920	0.940	0.980
Heart (Acc)	0.890	0.870	0.880	0.930	0.870	0.920	0.890	0.930
Diabetes (NMAE)	0.080	0.070	0.043	0.005	0.063	0.020	0.070	0.032

Detailed Results. Table 3 shows that all four models exhibit high recoverability across all datasets. TabICL achieves 95% accuracy on Adult at $K{=}0$, meaning the embedding reveals nearly all sensitive attributes without any side information. The Privacy Cliff generalizes across architectures: every model shows a $K{=}0 \rightarrow K{=}1$ jump (e.g., Mitra on German: 78% → 92%; TabDPT on Diabetes NMAE: 0.043 → 0.005). Interestingly, TabDPT, Mitra, and TabICL show higher $K{=}0$ recoverability than TabPFN, suggesting their embeddings are more structured and thus more immediately decodable, while TabPFN's more entangled representations produce a steeper cliff.

5.5 RQ4: Is Recoverability Merely Tabular Correlation?

Motivation. High recoverability could stem from correlations among tabular features rather than information encoded in the embedding. We compare Cascade with four baselines that use public features and reference data without the target embedding.

Table 4: Attribute recovery from public features or embeddings. Baselines use K public features without the target embedding. Cascade uses the embedding at $K{=}0$. Accuracy (↑) for Adult, German, and Heart, and NMAE (↓) for Diabetes. Best per row in bold.

	KNN		**GBDT**		**MICE**		**TabPFN-imputer**		**Cascade**
Dataset	$K{=}1$	$K{=}4$	$K{=}1$	$K{=}3$	$K{=}1$	$K{=}3$	$K{=}1$	$K{=}3$	$K{=}0$
Adult	0.200	0.370	0.448	0.495	0.253	0.250	0.448	0.502	**0.650**
German	0.380	0.440	0.417	0.387	0.185	0.189	0.419	0.434	**0.740**
Heart	0.610	0.590	0.525	0.565	0.384	0.448	0.544	0.609	**0.790**
Diabetes	0.160	0.100	0.121	0.098	0.106	0.091	0.105	0.085	**0.080**

Summary of Findings. Cascade at $K{=}0$ outperforms all four baselines across the reported settings (Table 4). This supports the value of embedding information beyond recovery from revealed features alone under the evaluated baselines.

Detailed Results. KNN uses five neighbors in the Shadow setting. Gradient-boosted decision trees (GBDT) predict sensitive attributes from revealed features, multiple imputation by chained equations (MICE) performs chained conditional imputation, and TabPFN-imputer uses TabPFN for attribute imputation. All four baselines use public features and reference data without the target embedding, whereas Cascade receives the embedding with no public features. Across the reported public-feature settings, Cascade exceeds the best non-embedding accuracy by 14.8 percentage points on Adult, 30.0 on German, and 18.0 on Heart. The gap is smaller on Diabetes, where Cascade achieves 0.080 NMAE compared with 0.085 for TabPFN-imputer at $K{=}3$.

5.6 RQ5: Can Embedding Transformations Mitigate Recoverability?

Motivation. A natural follow-up question is whether simple post-processing of embeddings before release can reduce recoverability while preserving their downstream utility. We evaluate three representative embedding-space transformations.

Summary of Findings. No transformation eliminates recoverability while preserving utility. Gaussian noise at moderate levels ($\sigma{=}0.5$) offers the best tradeoff, but still leaves substantial leakage. Knowledge distillation is counterproductive, actually increasing recoverability.

Table 5: Mitigation evaluation at K=0, S=all (Shadow, Cascade Probing). Rec is categorical recoverability (↓), and Util is downstream k-NN accuracy (↑). For Diabetes, Rec is NMAE (↑). Arrows indicate better mitigation or preserved utility.

	Adult		**German**		**Heart**		**Diabetes**	
Mitigation	Rec	Util	Rec	Util	Rec	Util	Rec	Util
None	0.932	0.870	0.758	0.703	0.812	0.800	0.056	0.675
Noise σ=0.1	0.839	0.855	0.693	0.723	0.753	0.800	0.067	0.675
Noise σ=0.5	0.679	0.860	0.589	0.733	0.629	0.833	0.101	0.675
Noise σ=1.0	0.607	0.855	0.547	0.723	0.567	0.800	0.118	0.675
Noise σ=5.0	0.449	0.780	0.409	0.663	0.436	0.733	0.151	0.636
PCA k=4	0.734	0.860	0.637	0.713	0.666	0.800	0.083	0.740
PCA k=8	0.830	0.865	0.668	0.723	0.710	0.800	0.068	0.701
PCA k=32	0.921	0.870	0.741	0.703	0.798	0.800	0.057	0.675
KD (64-dim)	0.953	0.870	0.807	0.713	0.858	0.767	0.023	0.727

Detailed Results. We evaluate three embedding-space mitigations (Table 5): Gaussian noise injection, PCA dimensionality reduction, and knowledge distillation (KD). Gaussian noise at σ=0.5 offers the best privacy/utility tradeoff, reducing recoverability by 17–25 percentage points with k-NN utility unchanged or slightly improved (Adult: −1pp; German and Heart: +3pp; Diabetes: 0). Aggressive noise (σ=5.0) reduces recoverability further but degrades utility by 4–9 percentage points. PCA k=4 provides moderate protection; $k\geq$32 has negligible effect. KD is counterproductive: the compressed student representation *increases* recoverability by 2–5 percentage points on categorical datasets (and reduces Diabetes NMAE from 0.056 to 0.023), as the distilled embeddings concentrate task-relevant information. No mitigation eliminates recoverability while preserving embedding utility.

5.7 Additional Analyses

We conduct extensive ablations and additional experiments in the Appendix. **Recovery order** (Appendix J.1): we compare random, reverse, greedy, and exhaustive orderings, confirming that our greedy heuristic is near-optimal. **Decoder architecture** (Appendix J.2): we test 7 architectures spanning a 292× parameter range (49K–14.3M); even a linear probe achieves comparable recoverability, confirming the leakage is intrinsic to the embeddings. **Per-feature cliff** (Appendix J.3): we rank which anchor features most increase recoverability, finding that *any* single feature triggers the cliff. **Iterative refinement** (Appendix J.4): a Joint-then-refine baseline improves over Joint but remains substantially below Cascade, confirming that sequential ordering, not just iteration, is what helps. **L2 normalization** (Appendix J.5): standard post-processing does not reduce recoverability. **Target normalization** (Appendix J.7): min-max normalization is critical for Joint but Cascade is robust without it. **High-dimensional datasets** (Appendix J.8). Cascade outperforms all evaluated baselines on COIL2000 and Communities & Crime when recovering 50 sensitive attributes. **Computational overhead** (Appendix J.9). Discovery becomes costlier with more sensitive targets, but its one-time cost is amortized across records and inference remains fast.

6 Conclusion

We presented a systematic privacy audit of TabFM embeddings as data-sharing artifacts. Our method, **Cascade Probing**, leverages the chain rule to sequentially recover sensitive attributes from embeddings. It reveals substantially higher leakage than joint probing and optimization-based inversion baselines, providing a more faithful assessment of the true privacy risk. Our central finding is the **Privacy Cliff**: a single known attribute sharply increases the recoverability of all remaining attributes. This phenomenon generalizes across four TabFM architectures (TabPFN, TabDPT, Mitra, TabICL) and multiple datasets. Our evaluation of standard embedding-space mitigations (noise injection, PCA, knowledge distillation) shows that none eliminates recoverability without significantly harming utility, indicating that post-hoc defenses are insufficient. These results show that TabFM embeddings encode highly entangled attribute information and should not be assumed safe for a privacy-preserving release without explicit auditing.

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Justification: The main results (Tables 1, 2) are averaged over all S (number of sensitive attributes). The recovery order ablation (Table 34, Appendix J.1) reports standard deviations ($\pm$ std) for random orderings over 5 permutations.

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   Justification: Appendix J.9 reports wall-clock times for Strategy Discovery and per-record inference on an NVIDIA A100 GPU, broken down by the number of sensitive attributes S.

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A Impact Statement and Limitations

This work reveals that TabFM embeddings, increasingly shared as substitutes for raw tabular records, retain highly recoverable attribute-level information. By introducing **Cascade Probing**, we show that inter-attribute correlations within embeddings can be systematically exploited to recover sensitive data, demonstrating that current embeddings are far more transparent than previously assumed. Our discovery of the "Privacy Cliff", where minimal side information triggers a sharp increase in recoverability, challenges the assumption that embeddings can be safely released or shared across institutional boundaries. Furthermore, our validation using LLM-synthesized reference data shows that effective auditing (and, by extension, potential misuse) requires minimal real data, lowering the practical barrier. These results underscore an urgent need for privacy-preserving mechanisms, such as differentially private embedding generation, to safeguard structured data in high-stakes domains. We release our audit framework to help data owners evaluate embedding privacy before deployment.

Limitations. This work focuses on diagnosing privacy leakage in TabFM embeddings but does not propose a defense that eliminates recoverability while preserving utility. Our evaluation of

post-hoc mitigations (noise, PCA, knowledge distillation) shows that none is sufficient, but designing principled privacy-preserving embedding mechanisms remains an open problem that requires further research. Additionally, the quality of LLM-synthesized reference data depends on the generator model and the informativeness of the schema metadata; domains with highly specialized or proprietary feature semantics may yield weaker synthetic data, limiting the applicability of the Few-Shot and Zero-Shot audit settings.

B Dataset Details

Table 6 summarizes the ten datasets used in our evaluation.

Table 6: Summary of evaluation datasets. Feature counts and types refer to the representations used in our experiments.

Dataset	Samples	Features	Cat.	Cont.	Domain
Adult Census	48,842	13	8	5	Demographics
German Credit	1,000	20	13	7	Finance
Heart Disease	303	13	8	5	Healthcare
Diabetes Pima	768	8	0	8	Healthcare
ACS Employment	50,000	16	16	0	Census
ACS Pub. Coverage	50,000	19	19	0	Census
ACS Inc.-to-Poverty	50,000	19	19	0	Census
ACS Health Ins.	50,000	25	25	0	Census
COIL2000	9,822	85	85	0	Insurance
Communities & Crime	1,994	100	0	100	Socioeconomic

Adult Census Income [1]: Predicts whether income exceeds \$50K/year based on census data. Contains demographic features (age, sex, race), employment features (workclass, occupation), and financial features (capital-gain/loss).

German Credit [15]: Credit risk assessment dataset with 20 features covering personal information (age, housing, job), account status (checking, savings), and loan details (amount, duration, purpose).

Heart Disease (Cleveland) [18]: Cardiac disease diagnosis with 13 clinical features including demographics (age, sex), symptoms (chest pain type), and medical measurements (blood pressure, cholesterol, heart rate).

Diabetes Pima [27]: Diabetes prediction for Pima Indian women with 8 continuous features: pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, and age.

ACS Employment [6]: Employment prediction from the US Census ACS PUMS 2018 (all 50 states + PR) via the folktables benchmark. Contains 16 categorical features covering demographics (age, sex, race, marital status, citizenship, nativity), disability indicators (hearing, vision, cognitive, ambulatory), military service, and household relationship. Subsampled to 50K.

ACS Public Coverage [6]: Public health insurance coverage prediction from the same ACS source. Contains 19 categorical features including the employment features above plus state (51 categories), income (binned into 7 brackets), employment status, and fertility. Subsampled to 50K.

ACS Income-to-Poverty Ratio [6]: Poverty prediction (income-to-poverty ratio $< 250\%$) from ACS. Contains 19 categorical features including demographics, disability indicators, employment status, weekly work hours, and grandchild care. Subsampled to 50K.

ACS Health Insurance [6]: Private health insurance prediction from ACS. Contains 25 categorical features, the largest feature set among the ACS datasets, covering demographics, disability, employment, household, and income variables. Subsampled to 50K.

COIL2000 [29] contains customer demographics and insurance-product information. Our evaluation uses 9,822 records and treats the 85 input features as categorical.

Communities & Crime [31] combines socioeconomic, law-enforcement, and crime statistics for US communities. Our evaluation uses 1,994 records and 100 continuous input features. Both wider

datasets are evaluated in the Shadow setting at $K{=}0$ with $S{=}50$ sensitive targets, with additional timing measurements at $S \in \{10, 30, 50\}$.

C Feature Priority Lists

To systematically evaluate recoverability across features of varying sensitivity and observability, we define feature priority lists for each dataset (see Tables 8, 10, 12 and 14). These priority lists establish the order in which features are considered during evaluation:

- **Public Priority**: Features ordered from most public (easily observable) to least public. These represent information a recipient could plausibly know.
- **Sensitive Priority**: Features ordered from most sensitive (highest privacy concern) to least sensitive. These represent the primary recovery targets.

The priority lists serve two purposes:

1. **Feature Selection Order**: When evaluating Cascade Probing, features are processed according to these priority orderings.
2. **Realistic Scenarios**: The orderings reflect real-world settings where recipients typically possess easily observable demographic information while sensitive financial or health data is held out.

Note that a feature may appear in both priority lists. For a given configuration (K, S), we first select the top-K features from the public list as x_{pub}, then select the top-S features from the sensitive list that are not already in x_{pub} as x_{sens} (skipping any overlap). This ensures the partition is always disjoint.

C.1 Adult Census Income

The Adult Census Income dataset (Table 7) contains 13 features for predicting whether income exceeds \$50K/year.

Table 7: Features in the Adult dataset.

Idx	Feature	Type
0	age	Integer
1	workclass	Categorical
2	education	Categorical
3	education-num	Integer
4	marital-status	Categorical
5	occupation	Categorical
6	relationship	Categorical
7	race	Categorical
8	sex	Binary
9	capital-gain	Integer
10	capital-loss	Integer
11	hours-per-week	Integer
12	native-country	Categorical

C.2 German Credit

The German Credit dataset (Table 9) contains 20 features for credit risk assessment.

C.3 Heart Disease

The Heart Disease (Cleveland) dataset (Table 11) contains 13 features for cardiac disease diagnosis.

Table 8: Priority lists for Adult dataset.

Public Priority		Sensitive Priority	
Rank	**Feature**	**Rank**	**Feature**
1	race	1	occupation
2	sex	2	age
3	native-country	3	education
4	education	4	marital-status
5	education-num	5	relationship
6	workclass	6	capital-gain
7	hours-per-week	7	capital-loss
8	capital-gain	8	hours-per-week
9	capital-loss	9	workclass
10	marital-status	10	education-num
11	occupation	11	native-country
12	relationship	12	race
13	age	13	sex

Table 9: Features in the German Credit dataset.

Idx	**Feature**	**Type**	**Idx**	**Feature**	**Type**
0	checking_account	Categorical	10	present_residence	Integer
1	duration	Integer	11	property	Categorical
2	credit_history	Categorical	12	age	Integer
3	purpose	Categorical	13	other_installments	Categorical
4	credit_amount	Integer	14	housing	Categorical
5	savings_account	Categorical	15	num_credits	Integer
6	employment_duration	Categorical	16	job	Categorical
7	installment_rate	Integer	17	num_dependents	Integer
8	personal_status_sex	Categorical	18	telephone	Binary
9	other_debtors	Categorical	19	foreign_worker	Binary

Table 10: Priority lists for German Credit dataset.

Public Priority		Sensitive Priority	
Rank	**Feature**	**Rank**	**Feature**
1	foreign_worker	1	credit_amount
2	age	2	checking_account
3	personal_status_sex	3	savings_account
4	housing	4	credit_history
5	job	5	num_credits
6	telephone	6	purpose
7	num_dependents	7	duration
8	present_residence	8	installment_rate
9	property	9	employment_duration
10	other_installments	10	other_debtors
11	other_debtors	11	other_installments
12	employment_duration	12	property
13	num_credits	13	present_residence
14	installment_rate	14	num_dependents
15	checking_account	15	job
16	savings_account	16	housing
17	credit_history	17	personal_status_sex
18	purpose	18	age
19	duration	19	telephone
20	credit_amount	20	foreign_worker

Table 11: Features in the Heart Disease dataset.

Idx	Feature	Type
0	age	Integer
1	sex	Binary
2	chest_pain_type	Categorical
3	resting_bp	Integer
4	cholesterol	Integer
5	fasting_blood_sugar	Binary
6	rest_ecg	Categorical
7	max_heart_rate	Integer
8	exercise_angina	Binary
9	oldpeak	Continuous
10	slope	Categorical
11	num_vessels	Categorical
12	thal	Categorical

Table 12: Priority lists for Heart Disease dataset.

Public Priority		Sensitive Priority	
Rank	Feature	Rank	Feature
1	sex	1	fasting_blood_sugar
2	age	2	exercise_angina
3	chest_pain_type	3	chest_pain_type
4	fasting_blood_sugar	4	slope
5	rest_ecg	5	thal
6	exercise_angina	6	num_vessels
7	slope	7	rest_ecg
8	thal	8	max_heart_rate
9	num_vessels	9	resting_bp
10	oldpeak	10	cholesterol
11	max_heart_rate	11	oldpeak
12	resting_bp	12	age
13	cholesterol	13	sex

C.4 Diabetes Pima

The Diabetes Pima dataset (Table 13) contains 8 features for diabetes prediction in Pima Indian women. All features are continuous.

Table 13: Features in the Diabetes Pima dataset.

Idx	Feature	Type	Range
0	pregnancies	Integer	0–17
1	glucose	Continuous	44–199 mg/dl
2	blood_pressure	Continuous	24–122 mmHg
3	skin_thickness	Continuous	7–99 mm
4	insulin	Continuous	14–846 μU/ml
5	bmi	Continuous	18.2–67.1 kg/m^2
6	diabetes_pedigree	Continuous	0.08–2.42
7	age	Integer	21–81

D Privacy Attacks and Attribute Inference

Several classes of privacy attacks have been studied in the ML literature. We review three relevant attack families and their applicability to the embedding-sharing setting.

Membership Inference tests whether a specific record was part of the model's training set [32]. It is widely used for auditing trained models but targets training-data memorization and yields only a

Table 14: Priority lists for Diabetes Pima dataset.

Public Priority		Sensitive Priority	
Rank	**Feature**	**Rank**	**Feature**
1	glucose	1	pregnancies
2	bmi	2	age
3	age	3	blood_pressure
4	pregnancies	4	glucose
5	diabetes_pedigree	5	skin_thickness
6	skin_thickness	6	insulin
7	blood_pressure	7	diabetes_pedigree
8	insulin	8	bmi

Algorithm 1 Joint Probing (Training)

Input: Training Dataset $\mathcal{D}_{\text{train}}$, Embedding Model $\mathcal{T}$, Sets $\mathcal{X}_{\text{pub}}, \mathcal{X}_{\text{sens}}$
Output: Multi-Task Decoder Φ_{joint}
Initialize: Multi-Task Decoder Φ_{joint} with $|\mathcal{X}_{\text{sens}}|$ output heads
Compute $Z^{(i)}_{\text{full}} \leftarrow \mathcal{T}(x^{(i)})$ for all $x^{(i)} \in \mathcal{D}_{\text{train}}$
for each batch $(x^{(i)}, x^{(i)}_{\text{sens}}) \in \mathcal{D}_{\text{train}}$ **do**
    Mask inputs: $X^{(i)}_{\text{base}} \leftarrow \text{Mask}(x^{(i)}, \text{keep} = \mathcal{X}_{\text{pub}})$
    Compute base embedding: $z^{(i)}_{\text{base}} \leftarrow \mathcal{T}(X^{(i)}_{\text{base}})$
    Compute signals: $\Delta^{(i)} \leftarrow Z^{(i)}_{\text{full}} - z^{(i)}_{\text{base}}$
    Predict: $\hat{y}^{(i)} \leftarrow \Phi_{\text{joint}}([\Delta^{(i)}; z^{(i)}_{\text{base}}])$
    Update Φ_{joint} to minimize $\sum_{f \in \mathcal{X}_{\text{sens}}} \mathcal{L}(\hat{y}^{(i)}_f, x^{(i)}_{\text{sens}}[f])$
end for
Return Φ_{joint}

single bit of information (member or not). TabFM embeddings are generated for query records at inference time; the privacy risk lies in what the embedding reveals about the individual record, not in whether that record was used for training.

Data Extraction recovers verbatim training samples from a model, as demonstrated for large language models [4] and diffusion models [5]. Like membership inference, data extraction targets the training set rather than inference-time query records. Tabular records are also structured and heterogeneous (mixed categorical and continuous features), making verbatim extraction less informative than recovering specific attribute values.

Attribute Inference recovers individual sensitive attribute values of a specific record from the model's output [8, 19, 25]. Given an embedding z and optionally some public attributes, it measures how accurately the remaining sensitive columns can be decoded. The recovered values (e.g., income, diagnosis, occupation) are concrete and actionable. Since TabFM embeddings represent individual inference-time records, and the risk of sharing them lies in the exposure of specific attribute values, attribute inference provides the most direct measure of privacy leakage in this setting.

E Algorithm Details

E.1 Joint Probing

Joint Probing serves as a baseline that predicts all sensitive attributes simultaneously using a multi-task decoder trained on embedding residuals. Unlike Cascade Probing, it does not exploit inter-attribute dependencies. Pseudocode is given in Algorithms 1 and 2.

E.2 Strategy Discovery

Strategy Discovery is the offline procedure that determines the recovery order π and trains the per-attribute decoders Φ. Given a reference dataset, it greedily selects the next attribute to recover

Algorithm 2 Joint Probing (Inference)

Input: Record Embedding z_{rec}, Public Features x_{pub}, Decoder Φ_{joint}, Model $\mathcal{T}$
Output: Recovered Sensitive Attributes $\hat{x}_{\text{sens}}$
Let query input $q \leftarrow x_{\text{pub}}$ (pad missing with NaN)
Compute base embedding: $z_{\text{base}} \leftarrow \mathcal{T}(q)$
Compute residual: $\Delta \leftarrow z_{\text{rec}} - z_{\text{base}}$
Predict all targets: $\hat{x}_{\text{sens}} \leftarrow \Phi_{\text{joint}}([\Delta; z_{\text{base}}])$
Return $\hat{x}_{\text{sens}}$

Algorithm 3 Strategy Discovery

Input: Reference Dataset $\mathcal{D}_{\text{shadow}} = \{\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}\}$, Embedding Model $\mathcal{T}$, Sets $\mathcal{X}_{\text{pub}}, \mathcal{X}_{\text{sens}}$
Output: Inference Order π, Ordered Decoders Φ
Initialize: $\mathcal{U} \leftarrow \mathcal{X}_{\text{sens}}$, $\mathcal{R} \leftarrow \mathcal{X}_{\text{pub}}$, $\pi \leftarrow []$, $\Phi \leftarrow []$
Compute $Z^{(i)}_{\text{full}} \leftarrow \mathcal{T}(x^{(i)})$ for all $x^{(i)} \in \mathcal{D}_{\text{shadow}}$
while $\mathcal{U} \neq \emptyset$ **do**
  $best_score \leftarrow -\infty$, $best_pair \leftarrow \text{null}$
  Mask inputs: $X^{(i)}_{\text{base}} \leftarrow \text{Mask}(x^{(i)}, \text{keep} = \mathcal{R})$
  Compute signals: $\Delta^{(i)} \leftarrow Z^{(i)}_{\text{full}} - \mathcal{T}(X^{(i)}_{\text{base}})$
  for each candidate feature $f \in \mathcal{U}$ **do**
    Optimize ϕ_f on $\mathcal{D}_{\text{train}}$ to minimize $\mathcal{L}(\phi_f([\Delta^{(i)}; \mathcal{T}(X^{(i)}_{\text{base}})]), x^{(i)}_f)$
    $score \leftarrow \text{Perf}(\phi_f, \mathcal{D}_{\text{val}})$
    if $score > best_score$ **then**
      $best_score \leftarrow score$, $best_pair \leftarrow (f, \phi_f)$
    end if
  end for
  $(f^*, \phi^*) \leftarrow best_pair$
  π.Append(f^*), Φ.Append(ϕ^*)
  $\mathcal{U} \leftarrow \mathcal{U} \setminus \{f^*\}$, $\mathcal{R} \leftarrow \mathcal{R} \cup \{f^*\}$
end while
Return π, Φ

by training a probe for every remaining candidate and picking the one with the best validation performance, then conditions subsequent steps on the ground-truth value of the selected attribute. Pseudocode is given in Algorithm 3.

E.3 Cascading Inference

At inference time, Cascading Inference applies the trained decoders sequentially in the order π. At each step, the auditor reconstructs the query from previously recovered attributes, recomputes the residual against the target embedding, and predicts the next sensitive value. Pseudocode is given in Algorithm 4.

F Implementation Details

This section provides the implementation details for reproducing our experiments.

F.1 Decoder Architecture

Both Joint and Cascade decoders use a shared backbone architecture:

- **Input**: Concatenated residual and base embedding $[\Delta; z_{\text{base}}]$ (dimension: $2 \times d_{\text{emb}}$, where $d_{\text{emb}} = 192$ for TabPFN v2.5)
- **Backbone**: 3-layer MLP with hidden dimensions [512, 256, 128], ReLU activations, and dropout ($p = 0.1$) between layers

Algorithm 4 Cascading Inference

Input: Record embedding z_{rec}, public features x_{pub}, strategy (π, Φ), model $\mathcal{T}$
Output: Recovered sensitive attributes $\hat{x}_{\text{sens}}$
Initialize context $\hat{\mathcal{R}} \leftarrow x_{\text{pub}}$
for $t = 1$ **to** $|\pi|$ **do**
  Construct query $q \leftarrow \hat{\mathcal{R}}$ (pad missing with NaN)
  $\Delta \leftarrow z_{\text{rec}} - \mathcal{T}(q)$ ▷ Residual
  $\hat{x}_{\text{sens}}[\pi_t] \leftarrow \Phi_t([\Delta;\, \mathcal{T}(q)])$ ▷ Predict
  $\hat{\mathcal{R}}[\pi_t] \leftarrow \hat{x}_{\text{sens}}[\pi_t]$ ▷ Update context
end for
Return $\hat{x}_{\text{sens}}$

- **Output Heads**:
  - Categorical features: Linear layer $\rightarrow$ Softmax (output dim = number of categories)
  - Continuous features: Linear layer $\rightarrow$ Tanh (output dim = 1, scaled to $[-1, 1]$)

For **Joint Probing**, a single backbone feeds into $|\mathcal{X}_{\text{sens}}|$ parallel output heads. For **Cascade Probing**, each step t uses an independent decoder ϕ_{π_t} with the same architecture.

F.2 Training Hyperparameters

Table 15: Training hyperparameters for decoder optimization.

Parameter	**Value**
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Batch size	64
Epochs	100 (with early stopping)
Early stopping patience	10 epochs
Train/Val split	80% / 20%
Loss Functions	
Categorical targets	Cross-Entropy
Continuous targets	MSE (on normalized values)

F.3 Data Splits and Evaluation Protocol

Shadow Setting. We split the original dataset into three disjoint sets: 40% for TabPFN support (not accessible to the auditor), 50% as the reference dataset $\mathcal{D}_{\text{shadow}}$ for training probing decoders, and 10% as held-out test records for evaluation. The reference dataset is further split 80/20 for decoder training and validation.

Few-Shot and Zero-Shot Settings. For these low-resource scenarios, we sample N_{shot} real records (30 for Few-Shot, 0 for Zero-Shot) to prompt the LLM. We generate 2000 synthetic samples using the LLM, split 80/20 for decoder training/validation. Evaluation uses the same held-out test records as the Shadow setting for fair comparison.

Evaluation. For each (K, S) configuration, we report mean accuracy (categorical) and mean NMAE (continuous) averaged over 3 random seeds.

G LLM Augmentation Prompts

This section presents the prompt templates used for LLM-based synthetic data generation. We employ dataset-specific schemas containing feature names, descriptions, value ranges, categories, and domain constraints.

G.1 Few-Shot Prompt Template

For few-shot augmentation ($N > 0$), we use stratified generation with real examples. The prompt combines chain-of-thought reasoning with a constraint on a categorical feature to ensure diverse coverage.

```
[System] You are generating synthetic tabular data with a SPECIFIC CONSTRAINT. Before
generating, THINK about what characteristics are typical for this constraint.

DATASET: {dataset_name} - {description}

FEATURES:
- {feature_1}: {description_1} ({type_1}, range: {min_1}-{max_1})
- {feature_2}: {description_2} (categories: {categories_2})
- ...

CONSTRAINT: All samples must have {stratify_feature} = ‘‘{value}’’

EXAMPLES (study the relationships for this {stratify_feature} value):
{fewshot_examples_json}

GENERATE {n_samples} samples that:
- ALL have {stratify_feature} = ‘‘{value}’’ (REQUIRED)
- Maintain realistic correlations between features
- Show diversity within the group
- Are unique and not copies of examples

Return ONLY a valid JSON array.
```

Figure 4: Few-shot prompt template for LLM-based data augmentation.

G.2 Zero-Shot Prompt Template

For zero-shot generation ($N = 0$), our prompt incorporates three key components to elicit diverse samples without real data:

1. **Synthetic Diverse Examples**: We programmatically generate 3 example records that demonstrate the expected diversity, with values sampled uniformly across valid ranges for each feature.
2. **Value Distribution Requirements**: Explicit per-feature constraints specifying that categorical features must use ALL categories, binary features must include BOTH values, and continuous features must span LOW, MEDIUM, and HIGH ranges.
3. **Domain Constraints**: Dataset-specific correlations (e.g., for healthcare data: “max heart rate decreases with age”, “exercise-induced angina correlates with ST depression”).

```
DATASET: {dataset_name} - {description}

FEATURES:
- {feature_1}: {description_1} ({type_1}, range: {min_1}-{max_1})
- ...

DOMAIN CONSTRAINTS:
- {constraint_1}
- {constraint_2}
- ...

===== CRITICAL: DIVERSITY REQUIREMENT =====
Your output MUST look like these DIVERSE examples:
{synthetic_examples_json}

===== MANDATORY VALUE DISTRIBUTION =====
- {feature_1}: MUST use ALL categories [{cats}] across samples
- {feature_2}: MUST vary from {min} to {max}, use LOW/MED/HIGH
- ...

Generate {n_samples} samples as a JSON array.
```

Figure 5: Zero-shot prompt template with synthetic examples and value distribution requirements.

G.3 Diversity Hints

We rotate through different diversity hints across generation batches:

- "Focus on samples from the lower end of numerical ranges."
- "Focus on samples from the upper end of numerical ranges."
- "Generate samples representing edge cases near boundaries."
- "Generate a diverse mix covering the full valid range."

G.4 LLM Configuration

All experiments use the following settings:

Table 16: LLM generation hyperparameters.

Parameter	Value
LLM Model	Qwen2.5-7B-Instruct
Temperature	$\tau = 0.9$
Nucleus sampling	top-$p = 0.95$
Top-k sampling	$k = 100$
Repetition penalty	1.1
Samples generated per call	10

H Complete Recoverability Results

This section presents complete recoverability results organized by side information level (K). For each K, we report results across the four primary benchmarks (Adult, German Credit, Heart Disease, Diabetes Pima) and three data regimes (Shadow, Few-Shot, Zero-Shot). Each table shows performance for varying numbers of sensitive features (S), with a mean row at the bottom for comparison with main paper figures.

Metrics: We report mean categorical accuracy (higher is better) for datasets with categorical features, and NMAE% (lower is better) for continuous-only features.

H.1 No-Context ($K = 0$)

In the no-context setting, no side information is available, and recoverability is measured solely from the embedding z with $x_{\text{pub}} = \emptyset$.

H.1.1 Accuracy Results at K=0

The following tables (Tables 17 to 19) report the mean categorical accuracy.

H.1.2 Continuous Feature Results (NMAE) at K=0

The following tables (Tables 20 to 23) report NMAE for continuous features in datasets with mixed feature types.

Key observations at K=0:

- **Cascade dramatically outperforms Joint**: On Adult, Cascade achieves 65% mean accuracy in Shadow setting while Joint only reaches 22%. On German Credit, Cascade (74%) significantly outperforms Joint (44%).
- **Consistent across all datasets**: Heart Disease shows similar patterns with Cascade achieving 79–89% vs Joint's 33–55%.
- **Zero-Shot maintains effectiveness**: Even without real training data, Cascade Probing achieves strong performance (88% on Heart Disease, 53% on Adult) using LLM-generated synthetic data.

Table 17: Adult dataset results with K=0 (zero known features).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.10	0.19	0.08	0.14	0.14	0.14
2	0.07	0.19	0.14	0.10	0.14	0.14
3	0.12	0.58	0.11	0.41	0.06	0.42
4	0.27	0.69	0.18	0.62	0.24	0.56
5	0.27	0.67	0.32	0.59	0.15	0.64
6	0.30	0.69	0.12	0.67	0.19	0.63
7	0.17	0.73	0.19	0.58	0.28	0.60
8	0.19	0.71	0.19	0.56	0.25	0.44
9	0.32	0.84	0.25	0.79	0.11	0.76
10	0.36	0.91	0.34	0.79	0.33	0.73
11	0.27	0.92	0.43	0.82	0.21	0.77
Mean	**0.22**	**0.65**	**0.21**	**0.55**	**0.19**	**0.53**

Mean categorical accuracy reported (higher is better).

Table 18: German Credit dataset results with K=0 (zero known features).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
2	0.36	0.36	0.36	0.36	0.36	0.26
3	0.46	0.76	0.46	0.44	0.21	0.52
4	0.40	0.80	0.34	0.82	0.36	0.54
5	0.51	0.78	0.33	0.58	0.22	0.50
6	0.42	0.80	0.45	0.70	0.12	0.61
7	0.41	0.78	0.27	0.49	0.26	0.60
8	0.32	0.78	0.32	0.51	0.19	0.58
9	0.39	0.67	0.32	0.64	0.17	0.41
10	0.41	0.75	0.20	0.54	0.15	0.45
11	0.37	0.75	0.33	0.68	0.25	0.43
12	0.47	0.74	0.48	0.65	0.21	0.44
13	0.49	0.73	0.24	0.51	0.21	0.43
14	0.45	0.72	0.22	0.53	0.12	0.42
15	0.41	0.75	0.22	0.64	0.13	0.45
16	0.46	0.74	0.49	0.67	0.24	0.47
17	0.56	0.73	0.21	0.64	0.21	0.49
18	0.53	0.73	0.33	0.60	0.21	0.46
Mean	**0.44**	**0.74**	**0.33**	**0.59**	**0.21**	**0.49**

Mean categorical accuracy reported (higher is better). $S = 1$ omitted (continuous only).

- **Cascade reduces NMAE significantly**: On Diabetes Pima, Cascade reduces NMAE from 0.15–0.22 to 0.08–0.09 across all settings. For continuous features in mixed datasets, Cascade achieves 3–6$\times$ lower NMAE than Joint on Adult (0.04 vs 0.13–0.15) and German Credit (0.07 vs 0.41–0.43).
- **Joint struggles without categorical context**: On German Credit at S=1–2 (pure continuous features), Joint NMAE exceeds 0.80, indicating near-random predictions. Cascade maintains robust performance (NMAE 0.02–0.16).

H.2 Single-Feature Context (K=1)

With a single known feature ($K = 1$), the recipient has minimal side information. This reveals the Privacy Cliff phenomenon: even a single known feature dramatically increases recoverability.

H.2.1 Accuracy Results at K=1

The following tables (Tables 24 to 26) report the mean categorical accuracy.

Table 19: Heart Disease dataset results with K=0 (zero known features).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.83	0.83	0.17	0.83	0.17	0.83
2	0.70	0.70	0.63	0.78	0.30	0.78
3	0.52	0.82	0.48	0.86	0.37	0.84
4	0.50	0.88	0.45	0.89	0.34	0.89
5	0.55	0.80	0.35	0.91	0.46	0.90
6	0.48	0.83	0.37	0.92	0.35	0.93
7	0.44	0.75	0.37	0.92	0.29	0.89
8	0.56	0.77	0.30	0.91	0.29	0.90
9	0.49	0.79	0.34	0.91	0.42	0.89
10	0.54	0.79	0.39	0.91	0.33	0.90
11	0.51	0.80	0.29	0.90	0.36	0.91
12	0.53	0.76	0.30	0.92	0.30	0.90
Mean	**0.55**	**0.79**	**0.37**	**0.89**	**0.33**	**0.88**

Mean categorical accuracy reported (higher is better).

Table 20: Adult dataset NMAE with K=0 (continuous features only).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
2	0.16	0.06	0.16	0.06	0.15	0.06
3	0.20	0.02	0.34	0.08	0.17	0.06
4	0.15	0.04	0.16	0.04	0.17	0.04
5	0.15	0.03	0.17	0.04	0.17	0.04
6	0.12	0.02	0.10	0.03	0.17	0.02
7	0.06	0.01	0.08	0.02	0.09	0.02
8	0.07	0.06	0.11	0.04	0.06	0.04
9	0.10	0.02	0.10	0.03	0.07	0.03
10	0.08	0.01	0.10	0.02	0.09	0.03
Mean	**0.12**	**0.03**	**0.15**	**0.04**	**0.13**	**0.04**

NMAE reported (lower is better). S=1 omitted (no continuous sensitive features).

Table 21: German Credit dataset NMAE with K=0 (continuous features only).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.11	0.10	0.84	0.16	0.82	0.14
2	0.11	0.04	0.83	0.02	0.98	0.02
3	0.10	0.02	0.13	0.02	0.29	0.02
4	0.11	0.02	0.51	0.01	0.25	0.02
5	0.18	0.03	0.47	0.07	0.37	0.04
6	0.11	0.03	0.25	0.02	0.40	0.04
7	0.14	0.03	0.37	0.10	0.44	0.03
8	0.21	0.08	0.19	0.12	0.29	0.12
9	0.23	0.09	0.21	0.09	0.21	0.16
10	0.17	0.08	0.32	0.10	0.20	0.14
Mean	**0.15**	**0.05**	**0.41**	**0.07**	**0.43**	**0.07**

NMAE reported (lower is better). Joint struggles at low S in Few-Shot/Zero-Shot without categorical context.

H.2.2 Continuous Feature Results (NMAE) at K=1

The following tables (Tables 27 to 30) report NMAE for continuous features in datasets with mixed feature types at K=1.

Key observations at K=1:

Table 22: Heart Disease dataset NMAE with K=0 (continuous features only).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
8	0.18	0.10	0.23	0.03	0.18	0.03
9	0.17	0.13	0.36	0.04	0.31	0.04
10	0.16	0.11	0.37	0.05	0.50	0.04
11	0.15	0.10	0.36	0.04	0.46	0.04
12	0.16	0.11	0.35	0.04	0.36	0.05
Mean	**0.16**	**0.11**	**0.34**	**0.04**	**0.36**	**0.04**

NMAE reported (lower is better).

Table 23: Diabetes Pima dataset NMAE with K=0 (all features are continuous).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.16	0.16	0.28	0.18	0.24	0.17
2	0.17	0.10	0.24	0.13	0.23	0.13
3	0.15	0.08	0.17	0.06	0.20	0.07
4	0.15	0.06	0.17	0.05	0.17	0.05
5	0.15	0.06	0.19	0.05	0.19	0.09
6	0.12	0.06	0.22	0.05	0.24	0.05
7	0.12	0.06	0.19	0.07	0.29	0.05
Mean	**0.15**	**0.08**	**0.21**	**0.08**	**0.22**	**0.09**

NMAE reported (lower is better).

Table 24: Adult dataset results with K=1 (single known feature).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.99	0.96	0.94	0.91	0.81	0.78
2	0.98	0.98	0.93	0.92	0.84	0.78
3	0.96	0.95	0.89	0.89	0.81	0.79
4	0.96	0.96	0.92	0.89	0.84	0.85
5	0.96	0.96	0.90	0.90	0.83	0.83
6	0.96	0.95	0.90	0.91	0.84	0.86
7	0.96	0.95	0.91	0.90	0.84	0.83
Mean	**0.97**	**0.96**	**0.91**	**0.90**	**0.83**	**0.82**

Mean categorical accuracy reported (higher is better).

Table 25: German Credit dataset results with K=1 (single known feature).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
2	1.00	0.96	1.00	1.00	0.99	1.00
3	0.84	0.78	0.86	0.83	0.72	0.67
4	0.84	0.77	0.84	0.86	0.67	0.62
5	0.85	0.79	0.83	0.83	0.71	0.64
6	0.81	0.80	0.83	0.86	0.72	0.72
7	0.81	0.79	0.85	0.84	0.71	0.68
Mean	**0.86**	**0.81**	**0.87**	**0.87**	**0.75**	**0.72**

Mean categorical accuracy reported (higher is better).

- **Dramatic improvement from K=0**: Joint Probing jumps from 19–55% to 75–97% accuracy with just one known feature, confirming the Privacy Cliff.
- **Both methods perform well**: With context, the gap between Joint and Cascade narrows—both achieve high accuracy.

Table 26: Heart Disease dataset results with K=1 (single known feature).

	Shadow		Few-Shot		Zero-Shot	
S	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.90	0.90	0.97	1.00	0.97	0.97
2	0.85	0.83	0.90	0.97	0.80	0.88
3	0.91	0.83	0.87	0.92	0.87	0.88
4	0.88	0.85	0.73	0.81	0.72	0.79
5	0.85	0.81	0.78	0.82	0.73	0.82
6	0.87	0.87	0.84	0.86	0.82	0.86
7	0.81	0.79	0.79	0.84	0.72	0.84
8	0.83	0.78	0.68	0.83	0.78	0.84
9	0.84	0.76	0.80	0.85	0.79	0.87
10	0.82	0.80	0.80	0.83	0.73	0.86
11	0.80	0.80	0.80	0.83	0.77	0.83
12	0.82	0.78	0.79	0.84	0.68	0.84
Mean	**0.85**	**0.82**	**0.81**	**0.87**	**0.78**	**0.86**

Mean categorical accuracy reported (higher is better).

Table 27: Adult dataset NMAE with K=1 (continuous features only).

	Shadow		Few-Shot		Zero-Shot	
S	Joint	Cascade	Joint	Cascade	Joint	Cascade
2	0.02	0.01	0.03	0.02	0.03	0.03
3	0.02	0.01	0.03	0.02	0.04	0.03
4	0.02	0.01	0.03	0.02	0.04	0.03
5	0.03	0.01	0.03	0.02	0.04	0.03
6	0.01	0.01	0.02	0.01	0.03	0.01
7	0.01	0.00	0.02	0.01	0.02	0.01
Mean	**0.02**	**0.01**	**0.03**	**0.02**	**0.03**	**0.02**

NMAE reported (lower is better). S=1 omitted (no continuous sensitive features).

Table 28: German Credit dataset NMAE with K=1 (continuous features only).

	Shadow		Few-Shot		Zero-Shot	
S	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.02	0.02	0.01	0.02	0.02	0.03
2	0.02	0.02	0.01	0.01	0.02	0.02
3	0.03	0.02	0.03	0.01	0.03	0.02
4	0.04	0.02	0.02	0.01	0.03	0.02
5	0.04	0.04	0.03	0.03	0.05	0.04
6	0.05	0.03	0.04	0.03	0.06	0.03
7	0.04	0.03	0.03	0.02	0.05	0.03
Mean	**0.03**	**0.03**	**0.03**	**0.02**	**0.04**	**0.03**

NMAE reported (lower is better).

- **Cascade maintains advantage in low-resource settings**: On Heart Disease Zero-Shot, Cascade (86%) outperforms Joint (78%) by 8 percentage points.

H.3 Per-Attribute Recovery

Tables 31 and 32 report per-attribute recovery in the Shadow setting at $K{=}0$ and $K{=}1$ across Adult, German Credit, Heart Disease, and Diabetes Pima. The Heart and Diabetes results use $S{=}6$ and are averaged over three seeds. Their revealed attributes at $K{=}1$ are sex and glucose, respectively.

Table 29: Heart Disease dataset NMAE with K=1 (continuous features only).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
8	0.09	0.12	0.12	0.05	0.13	0.05
9	0.13	0.13	0.16	0.07	0.16	0.07
10	0.12	0.11	0.13	0.07	0.18	0.07
11	0.12	0.11	0.11	0.07	0.12	0.08
12	0.12	0.11	0.13	0.07	0.13	0.08
Mean	**0.12**	**0.12**	**0.13**	**0.07**	**0.14**	**0.07**

NMAE reported (lower is better). S=1–7 omit continuous features in sensitive set.

Table 30: Diabetes Pima dataset NMAE with K=1 (all features are continuous).

S	Shadow		Few-Shot		Zero-Shot	
	Joint	Cascade	Joint	Cascade	Joint	Cascade
1	0.07	0.09	0.09	0.10	0.08	0.09
2	0.05	0.07	0.07	0.07	0.07	0.08
3	0.06	0.06	0.07	0.07	0.07	0.08
4	0.05	0.06	0.06	0.07	0.07	0.07
5	0.05	0.05	0.06	0.06	0.06	0.06
6	0.05	0.06	0.07	0.07	0.08	0.07
7	0.05	0.05	0.06	0.06	0.08	0.06
Mean	**0.05**	**0.06**	**0.07**	**0.07**	**0.07**	**0.07**

NMAE reported (lower is better).

Table 31: Per-attribute categorical recovery for Cascade Probing in the Shadow setting. Acc. is accuracy, Bal. is balanced accuracy, and F1 is macro-F1 (all higher is better). Classes gives the number of categories. Majority is the majority-class rate, providing an accuracy baseline.

			$K = 0$			$K = 1$		
Attribute	**Classes**	**Majority**	**Acc.**	**Bal.**	**F1**	**Acc.**	**Bal.**	**F1**
Adult occupation	15	0.18	0.50	0.43	0.41	0.93	0.86	0.86
Adult education	16	0.34	0.77	0.65	0.57	0.92	0.83	0.76
Adult marital status	7	0.45	0.62	0.38	0.33	0.96	0.85	0.85
Adult relationship	6	0.40	0.68	0.61	0.54	0.94	0.84	0.85
German checking account	4	0.38	0.98	0.98	0.97	0.98	0.98	0.98
German savings account	5	0.61	0.61	0.20	0.15	0.78	0.59	0.59
German credit history	5	0.55	0.80	0.60	0.61	0.82	0.64	0.64
German purpose	11	0.25	0.65	0.39	0.38	0.85	0.57	0.56
Heart fasting blood sugar	2	0.80	0.80	0.50	0.44	0.80	0.53	0.49
Heart thal	3	0.60	0.82	0.64	0.61	0.81	0.71	0.67
Heart chest pain type	4	0.52	0.87	0.77	0.76	0.84	0.72	0.71
Heart number of vessels	4	0.46	0.74	0.64	0.63	0.69	0.58	0.54
Heart exercise angina	2	0.63	0.78	0.74	0.75	0.88	0.85	0.86
Heart slope	3	0.54	0.81	0.66	0.61	0.81	0.60	0.60

I Target Normalization Ablation: Full Results

This section provides complete results for the target normalization ablation study across all four datasets (Table 33), complementing the German Credit results in Table 39.

These findings have important implications for practitioners:

- **Joint Probing requires careful preprocessing**: Target normalization is essential for balanced multi-task learning across heterogeneous feature types.

Table 32: Per-attribute NMAE (↓) for Cascade Probing on Diabetes Pima in the Shadow setting, with S=6 and three seeds. Glucose becomes public at K=1 and diabetes pedigree enters the sensitive set.

Attribute	$K = 0$	$K = 1$
Insulin	0.056	0.044
Glucose	0.023	Public
Age	0.040	0.031
Skin thickness	0.044	0.038
Blood pressure	0.051	0.038
Pregnancies	0.066	0.055
Diabetes pedigree	Not targeted	0.064

Table 33: Effect of target normalization on attribute inference across all datasets (K=3, S=3).

Dataset	**Method**	**Accuracy ↑**		**NMAE ↓**	
		With	Without	With	Without
Adult	Joint	0.883	0.185	0.033	0.016
	Cascade	0.888	0.628	0.021	0.017
German Credit	Joint	0.940	0.230	0.043	0.140
	Cascade	0.935	0.930	0.039	0.151
Heart Disease†	Joint	—	—	0.038	0.117
	Cascade	—	—	0.050	0.070
Diabetes†	Joint	—	—	0.042	0.104
	Cascade	—	—	0.045	0.037

†Only continuous sensitive features at S=3.

- **Cascade Probing avoids this trap**: independent per-feature heads prevent loss-domination, removing the need for normalization.

J Additional Analyses

J.1 Recovery Order Ablation

We compare four recovery orderings (Shadow, K=0, S=4): (1) **Greedy** (ours): select features by validation performance, (2) **Random**: average over 5 random permutations (± std), (3) **Reverse**: hard-to-easy, (4) **Exhaustive**: best of all $S!$ permutations (oracle upper bound).

Table 34: Recovery order ablation at K=0, S=4 (Shadow, Cascade).

Dataset	**Greedy**	**Random**	**Reverse**	**Exhaustive**
Heart (Acc ↑)	**0.880**	0.791 ±.032	0.746	0.896
Diabetes (NMAE ↓)	**0.060**	0.066 ±.003	0.071	0.048

The greedy heuristic is the best non-oracle strategy, achieving performance within 1–2% of the exhaustive oracle. Reverse (hard-to-easy) performs worst.

J.2 Decoder Architecture Ablation

We test 7 decoder architectures spanning a 292× parameter range to assess sensitivity to decoder capacity (Few-Shot, Cascade, K=0, averaged over S).

Recoverability is stable across small-to-medium decoders (49K–3.3M parameters). Even a 49K linear decoder achieves comparable accuracy to the 363K default MLP. Very large decoders (4.8M+) underperform, consistent with overfitting in the high-capacity regime. This confirms the leakage is an intrinsic property of the embeddings.

Table 35: Decoder architecture ablation (K=0, Few-Shot, Cascade).

Architecture	Params	Adult Acc ↑	German Acc ↑	Heart Acc ↑	Diabetes NMAE ↓
Linear	49K	.500	.630	.760	.067
MLP 2L	58K	.560	.610	.870	.074
MLP 3L (default)	363K	.550	.590	.890	.080
MLP 4L	1.1M	.520	.590	.880	.069
Transformer 4L	3.3M	.540	.600	.780	.063
Transformer 6L	4.8M	.420	.510	.740	.057
Transformer 8L	14.3M	.370	.360	.690	.114

J.3 Per-Feature Privacy Cliff Analysis

To test whether specific anchor features drive the Privacy Cliff, we use each feature as the sole known feature (K=1), probe all remaining features (S=N−1), and rank features by the recoverability they enable (Shadow, Joint).

Table 36: Best vs. worst anchor feature across datasets.

Dataset	Best Anchor	Score	Worst Anchor	Gap
Adult (Acc)	workclass	0.960	education	0.036
German (Acc)	property	0.800	foreign_worker	0.061
Heart (Acc)	resting_ecg	0.852	resting_bp	0.077
Diabetes (NMAE)	glucose	0.010	pregnancies	0.033

Even the worst anchor yields high recoverability (lowest categorical accuracy is 0.739 on German Credit), meaning revealing *any* single feature is far more dangerous than revealing none. The gap between best and worst anchor reaches 7.7pp on Heart Disease.

J.4 Iterative Refinement Baseline

We implement an iterative refinement (IR) baseline that predicts all features jointly, then refines over T rounds by feeding predictions back as context (Shadow, K=0).

Table 37: Iterative Refinement vs. Joint and Cascade Probing (K=0, Shadow).

Dataset	Joint	IR (T=3)	IR (T=5)	Cascade
Heart Disease (Acc ↑)	0.550	0.610	0.620	**0.790**
Diabetes Pima (NMAE ↓)	0.150	0.120	0.110	**0.080**

IR improves over Joint, confirming that iterative correction helps. Cascade still outperforms IR by a large margin, showing that sequential dependency modeling is more effective than simultaneous prediction with refinement.

J.5 L2 Normalization Robustness

We test whether L2 normalization of embeddings reduces recoverability (Few-Shot, Cascade Probing, K=0, averaged over S).

Table 38: Effect of L2 normalization on probing performance.

Dataset	Baseline	L2-Norm
Adult (Acc)	.550	.540
German (Acc)	.590	.590
Heart (Acc)	.890	.910
Diabetes (NMAE)	.080	.080

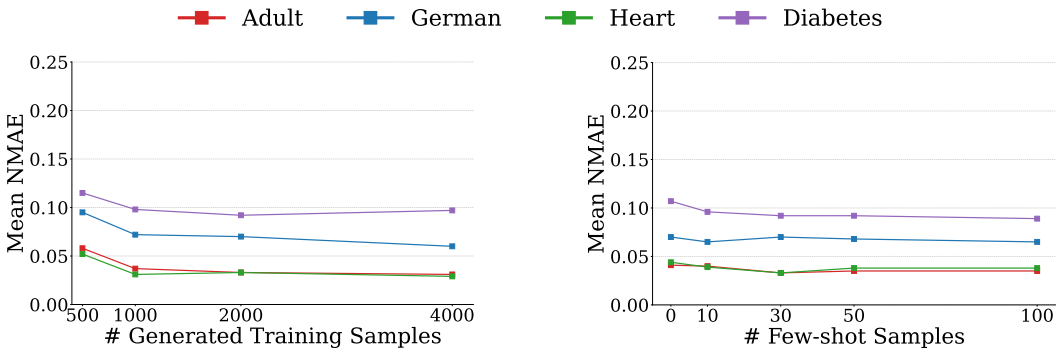
Table 39: Effect of target normalization on German Credit (K=3, S=3).

Method	Accuracy ↑		NMAE ↓	
	With	Without	With	Without
Joint	0.940	0.230	0.043	0.140
Cascade	0.935	0.930	0.039	0.151

L2 normalization does not reduce recoverability; performance is unchanged across all datasets.

J.6 LLM Data Efficiency

We study the efficiency of LLM-based shadow data generation by varying (i) the synthetic training budget N_{syn} and (ii) the few-shot prompting budget N_{shot} (K=0, Cascade, continuous features).



(a) Varying N_{syn}. (b) Varying N_{shot}.

Figure 6: LLM data efficiency (K=0, Cascade, mean NMAE on continuous features).

Increasing N_{syn} improves inversion with diminishing returns, flattening around 1000 samples. For the few-shot ablation (N_{syn}=2000), most improvement comes from the first few real exemplars.

J.7 Target Normalization Ablation

Target normalization is essential for Joint Probing. Table 39 shows that omitting normalization on German Credit collapses Joint categorical accuracy from 94.0% to 23.0%, while Cascade maintains robust accuracy (∼93%) even without normalization. Full results across all datasets are in Appendix I.

J.8 Recovery on High-Dimensional Datasets

We evaluate recovery of S=50 sensitive attributes on COIL2000 and Communities & Crime at K=0 in the Shadow setting. These experiments extend the evaluation to tables with 85 categorical and 100 continuous features, respectively. Unlike Table 1, which averages over sensitive-set sizes, Table 40 reports a fixed S=50.

Table 40: Recovery on high-dimensional datasets with S=50 sensitive targets at K=0 in the Shadow setting. COIL2000 reports categorical accuracy (↑) and Communities & Crime (C&C) reports NMAE (↓), averaged over the sensitive attributes. Best per row in bold.

Dataset (metric)	Joint	DirectOpt	WGAN	TabSyn	UIFV	Cascade
COIL2000 (Acc.)	0.674	0.346	0.411	0.443	0.430	**0.699**
C&C (NMAE)	0.064	0.145	0.108	0.099	0.094	**0.060**

Cascade achieves the strongest recovery on both datasets. Joint Probing is the closest baseline, with 0.674 accuracy on COIL2000 and 0.064 NMAE on Communities & Crime. The smaller gaps show that Cascade's advantage varies across datasets, while substantial recoverability persists with 50 sensitive targets. The cost of discovery and inference for these high-dimensional datasets is reported in Table 42.

J.9 Computational Overhead

Strategy Discovery is performed once for a fixed audit configuration. For S sensitive targets, it trains $S + (S-1) + \cdots + 1 = S(S+1)/2$ candidate probes. The selected probes and recovery order are retained and reused across records without retraining. We report wall-clock times on an NVIDIA A100. Table 41 reports the original German Credit experiment with 20 features and 2K training samples.

Table 41: Computational overhead of Strategy Discovery and per-record inference.

S	**Discovery time**	**Per-record inference**
1	5.5s	0.8ms
4	34.7s	1.8ms
10	3.7 min	3.9ms
20 (all)	12.0 min	7.2ms

Table 42 extends the timing evaluation to COIL2000 and Communities & Crime at $K{=}0$ in the Shadow setting. Discovery cost increases substantially with the number of sensitive targets, reaching 105.6 minutes on COIL2000 at $S{=}50$. This offline cost is amortized across records under the same audit configuration. Per-record inference grows approximately linearly over the measured settings and takes 13.1 ms on Communities & Crime and 36.9 ms on COIL2000 at $S{=}50$.

Table 42: Strategy Discovery and per-record inference times on high-dimensional datasets at $K{=}0$ in the Shadow setting, measured on an NVIDIA A100. Discovery is an offline cost for a fixed audit configuration.

	Communities & Crime		**COIL2000**	
S	Discovery (min)	Inference (ms/record)	Discovery (min)	Inference (ms/record)
10	1.0	2.6	4.5	7.6
30	6.0	7.8	37.7	22.7
50	17.1	13.1	105.6	36.9